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# Department of CSE

B.Tech. CSE Trimester -5

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# Computer Networks and Network Programming

**25CS2208E**

**LAB RECORD**

**2026**

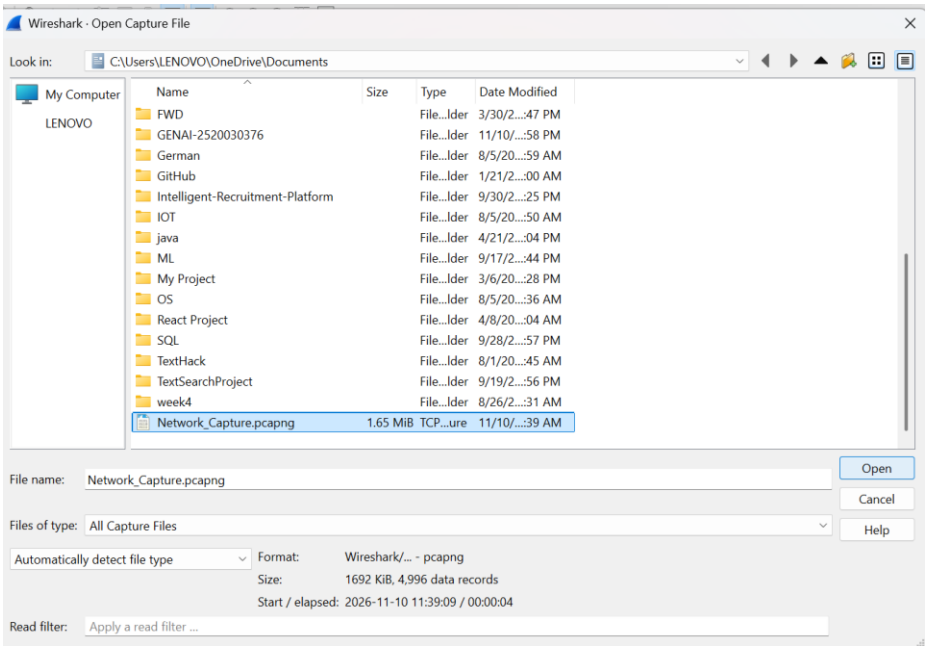
# Experiment -1

**AIM: To analyze a PCAP (Packet Capture) file using Wireshark, identify the total number of captured packets, determine the number of TCP, UDP, and ICMP packets using display filters, and identify the top five protocols using the Protocol Hierarchy statistics.**

## Steps:

### Step 1:Open the PCAP file

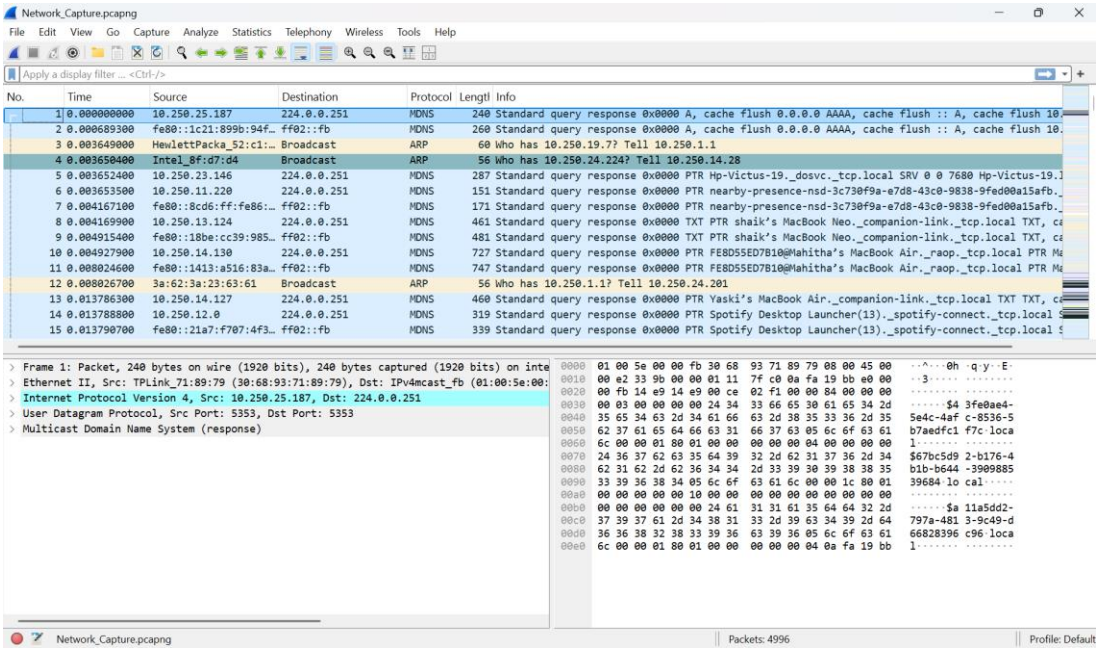
1. Open Wireshark.
2. Click on File → Open.
3. Select the required PCAP file.
4. Click Open to load the captured packets.
5. Observe the packets displayed in the Wireshark window.




### Step 2: Identify total number of packets

1. Open the PCAP file in Wireshark.
2. Look at the bottom of the Wireshark window.
3. Find the packet count displayed in the status bar.
4. Record the total number of packets.

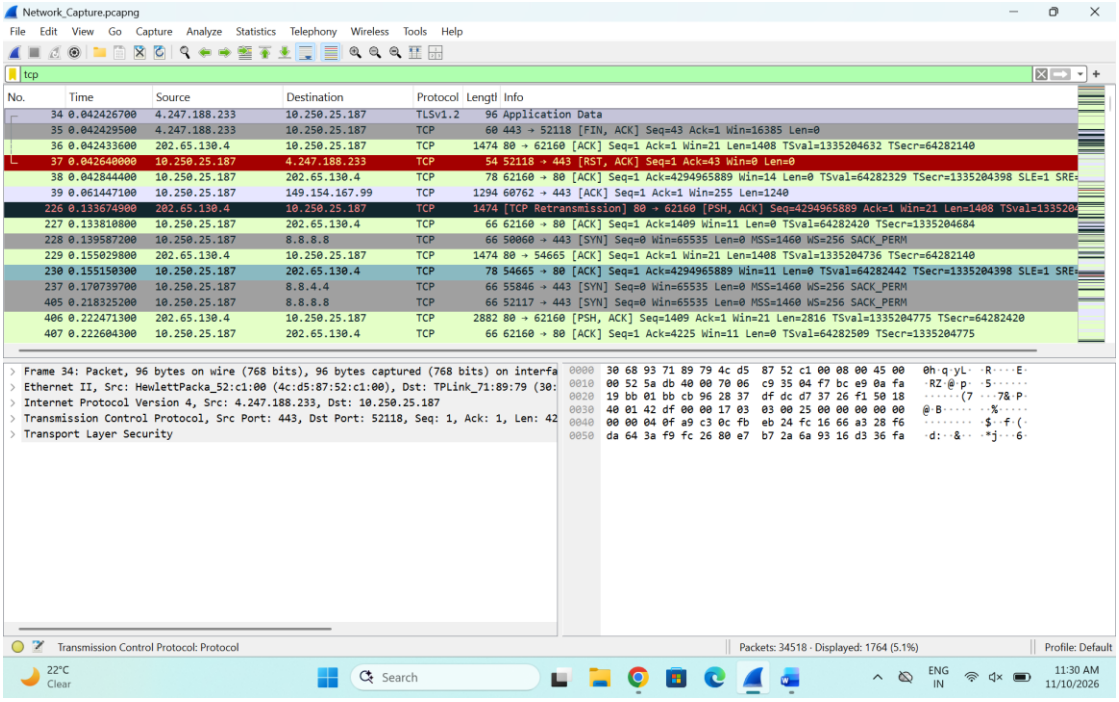
Result: Total packets captured = 4996




## Step 3: Identify TCP packets

1. Click the display filter bar at the top.
2. Type the filter:tcp
3. Press Enter.
4. Wireshark displays only TCP packets.
5. Check the packet count at the bottom.
6. Record the number.

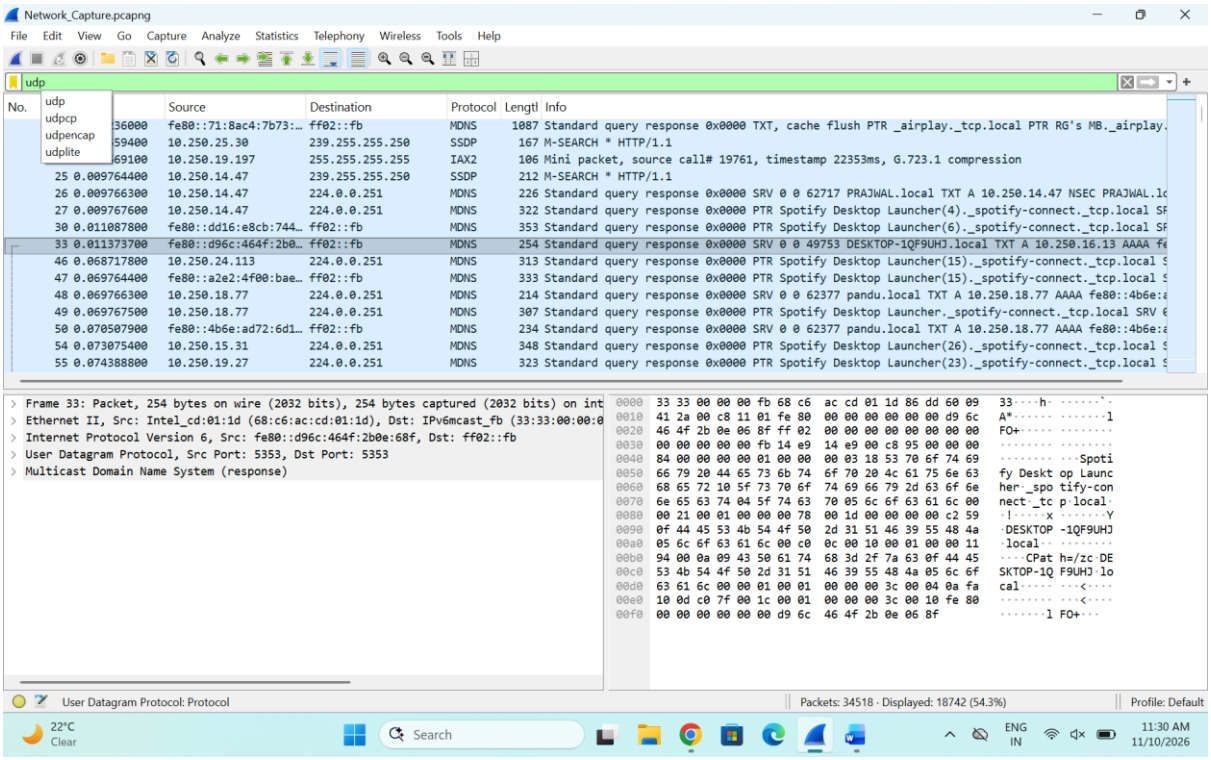
   Result: Total TCP packets = 34518, Displayed = 1764




## Step 4: Identify UDP packets

1. Click the display filter bar.
2. Remove the previous filter.
3. Type:udp
4. Press Enter.
5. Only UDP packets will be displayed.
6. Check the packet count at the bottom.
7. Record the number.

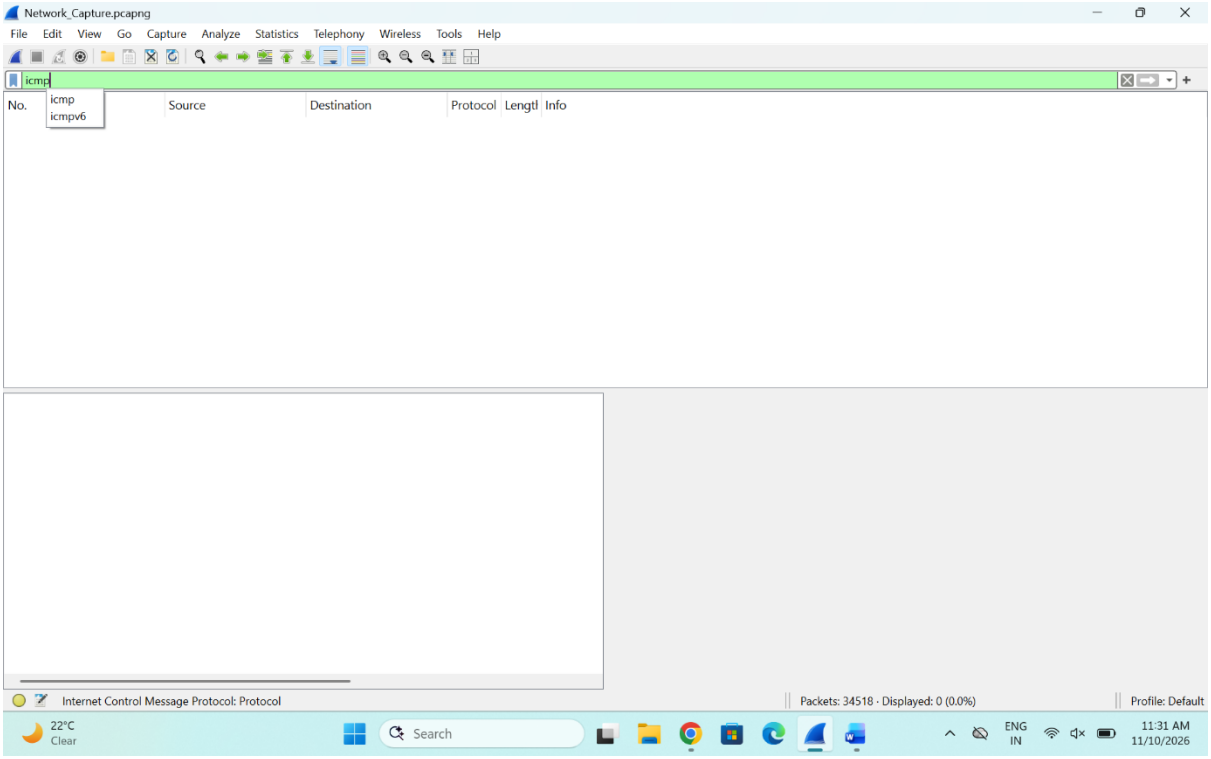
   Result: Total UDP packets = 34518, Displayed = 18742




## Step 5: Identify ICMP packets

1. Click the display filter bar.
2. Remove the previous filter.
3. Type:icmp
4. Press Enter.
5. Only ICMP packets will be displayed.
6. Check the packet count at the bottom.
7. Record the number.

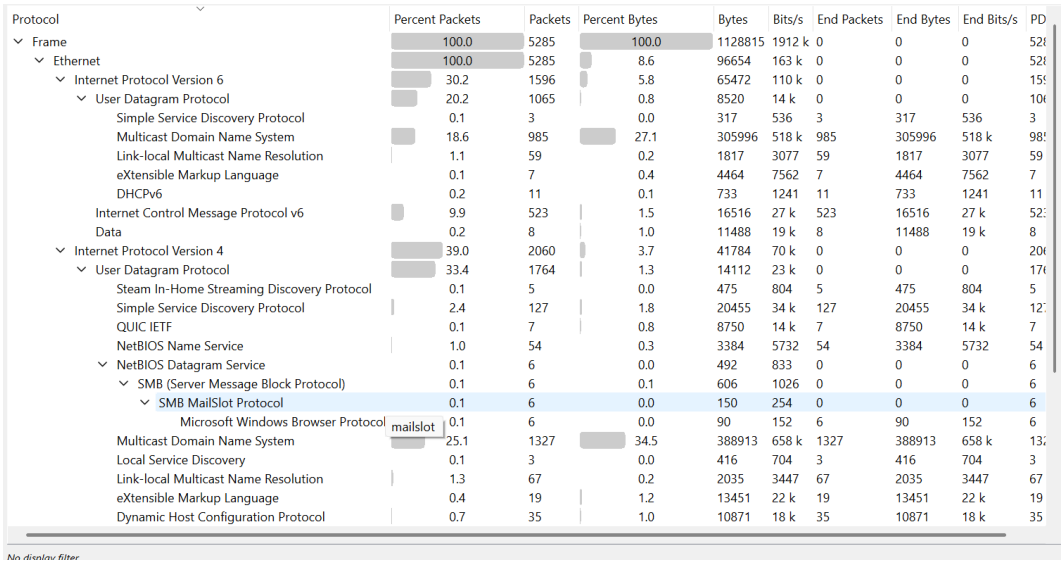
Result: Total ICMP packets = 34518 , Displayed = 0




# Step 6: Identify the top five protocols

1. Click on Statistics in the menu bar.
2. Select Protocol Hierarchy.
3. A new window will appear.
4. Observe the list of protocols.
5. Identify the top five protocols based on packet count or percentage.
6. Record their names and corresponding packet counts.
7. Take a screenshot for your record.

| Protocol | Packet Count |
|---|---|
| Multicast DNS (mDNS) | 2312 |
| ICMPv6 | 523 |
| Simple Service Discovery Protocol (SSDP) | 130 |
| Link-Local Multicast Name Resolution (LLMNR) | 126 |
| NetBIOS Name Service | 54 |



| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PD |
|---|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 5285 | 100.0 | 1128815 | 1912 k | 0 | 0 | 0 | 52 |
| Ethernet | 100.0 | 5285 | 8.6 | 96654 | 163 k | 0 | 0 | 0 | 52 |
| Internet Protocol Version 6 | 30.2 | 1596 | 5.8 | 65472 | 110 k | 0 | 0 | 0 | 15 |
| User Datagram Protocol | 20.2 | 1065 | 0.8 | 8520 | 14 k | 0 | 0 | 0 | 10 |
| Simple Service Discovery Protocol | 0.1 | 3 | 0.0 | 317 | 536 | 3 | 317 | 536 | 3 |
| Multicast Domain Name System | 18.6 | 985 | 27.1 | 305996 | 518 k | 985 | 305996 | 518 k | 98 |
| Link-local Multicast Name Resolution | 1.1 | 59 | 0.2 | 1817 | 3077 | 59 | 1817 | 3077 | 59 |
| eXtensible Markup Language | 0.1 | 7 | 0.4 | 4464 | 7562 | 7 | 4464 | 7562 | 7 |
| DHCPv6 | 0.2 | 11 | 0.1 | 733 | 1241 | 11 | 733 | 1241 | 11 |
| Internet Control Message Protocol v6 | 9.9 | 523 | 1.5 | 16516 | 27 k | 523 | 16516 | 27 k | 52 |
| Data | 0.2 | 8 | 1.0 | 11488 | 19 k | 8 | 11488 | 19 k | 8 |
| Internet Protocol Version 4 | 39.0 | 2060 | 3.7 | 41784 | 70 k | 0 | 0 | 0 | 20 |
| User Datagram Protocol | 33.4 | 1764 | 1.3 | 14112 | 23 k | 0 | 0 | 0 | 17 |
| Steam In-Home Streaming Discovery Protocol | 0.1 | 5 | 0.0 | 475 | 804 | 5 | 475 | 804 | 5 |
| Simple Service Discovery Protocol | 2.4 | 127 | 1.8 | 20455 | 34 k | 127 | 20455 | 34 k | 12 |
| QUIC IETF | 0.1 | 7 | 0.8 | 8750 | 14 k | 7 | 8750 | 14 k | 7 |
| NetBIOS Name Service | 1.0 | 54 | 0.3 | 3384 | 5732 | 54 | 3384 | 5732 | 54 |
| NetBIOS Datagram Service | 0.1 | 6 | 0.0 | 492 | 833 | 0 | 0 | 0 | 6 |
| SMB (Server Message Block Protocol) | 0.1 | 6 | 0.1 | 606 | 1026 | 0 | 0 | 0 | 6 |
| SMB MailSlot Protocol | 0.1 | 6 | 0.0 | 150 | 254 | 0 | 0 | 0 | 6 |
| Microsoft Windows Browser Protocol | 0.1 | 6 | 0.0 | 90 | 152 | 6 | 90 | 152 | 6 |
| Multicast Domain Name System | 25.1 | 1327 | 34.5 | 388913 | 658 k | 1327 | 388913 | 658 k | 13 |
| Local Service Discovery | 0.1 | 3 | 0.0 | 416 | 704 | 3 | 416 | 704 | 3 |
| Link-local Multicast Name Resolution | 1.3 | 67 | 0.2 | 2035 | 3447 | 67 | 2035 | 3447 | 67 |
| eXtensible Markup Language | 0.4 | 19 | 1.2 | 13451 | 22 k | 19 | 13451 | 22 k | 19 |
| Dynamic Host Configuration Protocol | 0.7 | 35 | 1.0 | 10871 | 18 k | 35 | 10871 | 18 k | 35 |